Non-local binding and person-blocking effects in Turkish

Metehan Oğuz & Elsi Kaiser (University of Southern California, moguz@usc.edu)

Understanding how syntactic and non-syntactic factors guide reflexive resolution informs how information is integrated in language comprehension. But many phenomena and languages are under-investigated, which poses risks of overlooking crosslinguistic similarities and conversely also not noticing typological variation. To broaden the empirical basis for theory-building, we (i) draw new crosslinguistic comparisons and (ii) test an under-researched structure in Turkish.

We investigate two reflexive pronouns in Turkish, to see if Blocking Effects occur in Turkish. In Chinese, an intervening 1st/2nd person pronoun blocks *ziji* 'self' in an embedded clause from long-distance binding [1,2]. This is often attributed to 1st/2nd person introducing a perspective center that blocks logophoric *ziji* from reaching the non-local antecedent (e.g. [1,3]).

Turkish has simple (*kendi*) and complex reflexives (*kendi-si*). Both can be bound by non-local antecedents (ex.1) [4,5], but whether *kendi* and *kendi-si* are perspective-sensitive logophors is not yet settled [6,7,8]. We test if these forms exhibit Chinese-style blocking effects in the presence of an interfering 1st person pronoun. If introduction of a perspectival center by a 1st person pronoun blocks non-local binding of *kendi* and *kendisi*, this would (a) provide novel evidence for logophoricity and (b) establish a new crosslinguistic generalization regarding blocking.

We also expand to a clause type whose binding patterns are under-researched: clauses headed by the complementizer *ki* (roughly, somewhat akin to English *that*). Prior work on non-local binding of *kendi* and *kendi-si* largely tested nominalized and finite embedded clauses (ex.1, [5,9]). Prior work argues that *ki*-clauses (analyzed as subordinate clauses [10,11] or paratactic structures with main clause properties [12]) are less permissive for cross-clausal operations, and may not allow long-distance dependencies. Thus, *ki*-clauses provide a more rigorous test of whether *kendi/kendi-si* can be bound non-locally.

Experiment (N_{SUBJ}=63, N_{ITEM}=24): We test what binding patterns are acceptable with *ki*-clauses; if there are blocking effects; and if the two reflexives pattern alike. We probe corerefence by asking people to rate if the critical words (reflexive and target antecedent, **bolded+boxed**), can refer to the same person (ex.2-3, see [13] for a methodology review). On a trial, the reflexive and one of the antecedents ('target': local sub, non-local sub, non-local obj) was bolded/boxed. We also manipulate the intervening NP (3rd/1st), and anaphor type (*kendi*/*kendisi*) (Table 1).

Results. When the target is in local subject position: *Kendi/kendisi* are 3rd person, so as expected coreference with a local 1st person subject is less acceptable (person mismatch) than with 3rd person (p<.001). We also find form-specific effects: *kendi* is more acceptable with local antecedents than *kendisi* (p<.001), similar to other clause types [e.g.4,5], and the cost of person mismatch is higher for *kendisi* (interaction, p<.01). **When the target is the non-local subject or object**, we find effects of antecedent position, intervenor type, and anaphor type (p's <.01): both reflexives are rated higher when bound by non-local subjects vs. objects and rated lower when the intervening noun is first-person, and *kendisi* is rated higher with non-local nouns than *kendi*.

As Fig.2 shows, ***kendi* exhibits strong blocking effects:** When the local subject is 1st person, non-local antecedents are rated very low (blocked). This is the first demonstration of person-blocking effects in Turkish. But with *kendisi* (Fig.1), presence of a 1st person local subject does **not block long-distance binding:** If anything, the non-local object becomes more available as an antecedent. Thus, *kendi* and *kendisi* pattern **asymmetrically** in terms of blocking (3-way interaction between anaphor antecedent position, intervenor type and anaphor type, p<.001).

In sum, long-distance binding of *kendi/kendisi* exists in *ki*-clauses, i.e. extends even to a context where cross-clausal dependencies are otherwise more constrained. Further, we provide the first evidence that *kendi* is subject to person blocking, and that first-person blocking exists in Turkish. If we view blocking as perspectival, this provide new evidence of *kendi* being perspective-sensitive. An intriguing question is posed by the asymmetrical behavior of *kendisi*. As a whole, we provide further evidence for form-specific accounts [e.g.12,13].

(1) *Subordinate (embedded) clause is nominalized (configuration tested in prior work, [5,9]):*
Ali$_1$ Can-a$_2$ [Murat-ın$_3$ {*kendi/kendisi*}$_{1/2}$-ni sev-diğ-i-ni] söyle-di.
Ali Can-DAT Murat-GEN self-ACC like-NMLZ-3SG-ACC tell-PST
"Ali$_1$ told Can$_2$ that Murat$_3$ likes *him*$_{1/2/3}$."

(2) *Ki-clause structure*
Ali$_1$ Can-a$_2$ de-di **ki** [{**Murat/ben**}$_3$ maç-ta {*kendi/kendisi*}$_{1/2/3}$-ni suçla-dı-{Ø/m}].
Ali Can-DAT say-PST **KI** **Murat/1SG** game-LOC self-ACC blame-PST-3SG-1SG
"**Ali$_1$** told **Can$_2$** **that** **Murat$_3$** blamed *self*$_{1/2/\#3}$ at the game."

(3) Screenshot (How acceptable is the sentence when two **bold/boxed words** refer to the same person? 1 = very unacceptable, 5= very acceptable)

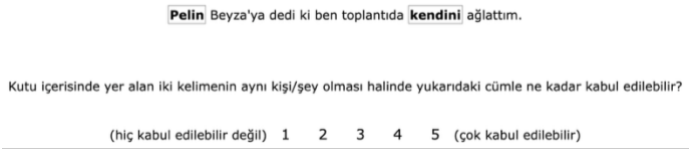

Table 1. Design and example item (All targets used 'told', i.e. the matrix subject is a source)

Target/ binder	*Emb. sub.*	Anaphor type (*kendi*/*kendisi*) was manipulated between-subjects
Local	3rd	Ali$_1$ Can-a$_2$ de-di ki [**Murat**$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-Ø]. Ali told Can that **Murat**$_{LOCAL}$ blamed **himself** at the game.
Non-loc sub	3rd	**Ali**$_1$ Can-a$_2$ de-di ki [Murat$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-Ø]. **Ali**$_{SUBJ}$ told Can that Murat blamed **himself** at the game.
Non-loc obj	3rd	Ali$_1$ **Can-a**$_2$ de-di ki [Murat$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-Ø]. Ali told **Can**$_{OBJ}$ that Murat blamed **himself** at the game.
Local	1st	Ali$_1$ Can-a$_2$ de-di ki [**ben**$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-m]. Ali told Can that **I**$_{LOCAL}$ blamed **himself** at the game.
Non-loc sub	1st	**Ali**$_1$ Can-a$_2$ de-di ki [ben$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-m]. **Ali**$_{SUBJ}$ told Can that I blamed **himself** at the game.
Non-loc obj	1st	Ali$_1$ **Can-a**$_2$ de-di ki [ben$_3$ maç-ta **{kendi/kendisi}**$_{1/2/3}$-ni suçla-dı-m]. Ali told **Can**$_{OBJ}$ that I blamed **himself** at the game.

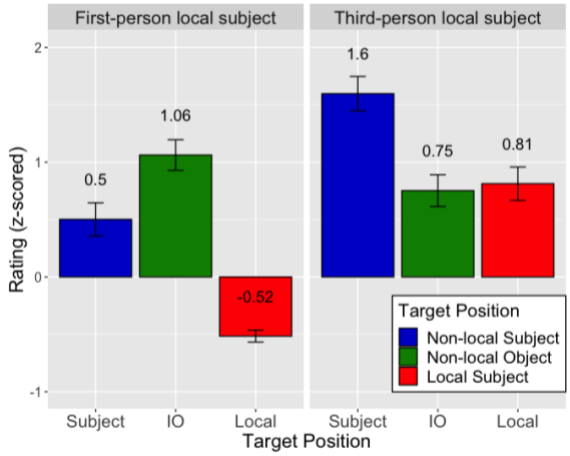

Fig.1. Mean acceptability ratings for kendisi

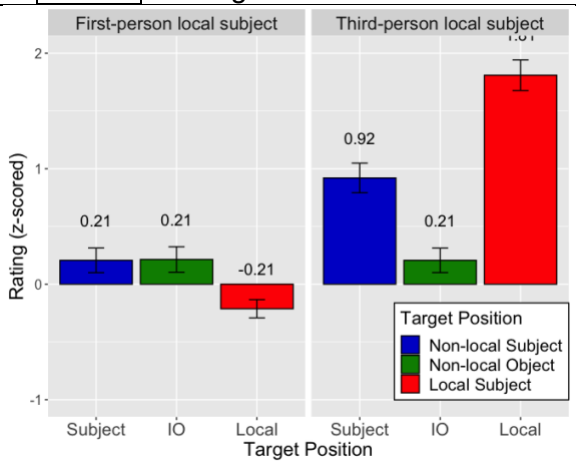

Fig.2 Mean acceptability ratings for kendi

References [1] Huang et al 1984 *WCCL* [2] Xue et al 1994 *WCCFL* [3] Charnavel et al 2017 *Long-distance anaphora* [4] Özbek & Kahraman 2016 Mersin Uni. Dil Edebiyat [5] Gračanin-Yüksek et al 2017 *JPR* [6] İşsever 2015 *TU+* [7] Sezer 1980 Harvard Ukrainian Studies [8] Kornfilt 2001 *LDRs, Syntax, Semantics* [9] Oğuz & Kaiser 2023 *TU+8* [10] Kornfilt 1997 *Turkish* [11] Göksel & Kerslake 2005 *Turkish*. [12] Predolac 2017 *Syntax of complementation in Turkish* [13] Kaiser 2021 Anaphora. [14] Kaiser & Trueswell 2008 *LCP* [15] Kaiser et al 2009 *Cogn*.